

Predicting Revenue by Yellow Taxis in New York City

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Github repo with commit

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1 Introduction

Yellow taxi drivers operating in New York City face a critical decision: where should they work on any given day to maximize revenue? Yet they lack tools to quantify how competition, demand patterns, and geographic factors influence earnings in different zones. This paper develops predictive models to answer this question, enabling drivers to make location decisions based on data rather than intuition.

We start by defining yellow taxi **revenue** as the total amount of money charged to passengers (`total_amount`, not including cash tips), as defined by the New York City Taxi and Limousine Commission (NYC TLC) [7].

Since taxi drivers would be interested in knowing which taxi zones to travel to for maximising revenue, we analysed the data at a **daily frequency** and ensured that our models are able to predict revenue earned on a particular zone for a particular day.

We chose the time period of 2023-2026 as this comes after the disruptions of the COVID-19 pandemic and thus is well past any possible disruptions from events such as lockdowns or tightly controlled usage of airports. We also try to use data that is as recent as possible as this would benefit taxi drivers with data that is maximally relevant to the present. A 3.5 year time period was chosen as it gives enough data to work with for detecting seasonal patterns in earnings that are only observable over a long time horizon.

2 Datasets

Dataset	Records	Columns
TLC Yellow Taxi Trip Record Data [7]	147201830	20
TLC FHVHV Trip Record Data [7]	821546266	25
TLC Taxi Zone Lookup Table [5]	265	4
TLC Taxi Zone Shapefile [6]	263	7
MapPLUTO CSV file [4]	856688	102
MapPLUTO Shapefile [4]	N/A	N/A
MTA Train Stations Data [3]	496	19

Table 1: Shapes of all datasets used

3 Methodology

We performed the following sequence of steps to conduct our analysis.

1. Acquired all datasets, including NYC yellow taxi and high volume for-hire (HVFHV) trip data over the timeframe of January 2023 to end of May 2026.
2. Removed all records that had anomalous values for our features of interest (the record removal and imputation rules are listed in the Preprocessing section below).
3. Aggregated the data to a daily frequency and calculated daily total amount for each combination of taxi zone ID (`PULocationID`) and pick up date (`tpep_pickup_datetime`). This value was then log transformed after we observed a significantly right-skewed distribution.
4. Performed similar preprocessing steps to Step 2 for the HVFHV dataset and then calculated trip volume as the number of instances belonging to each combination of taxi zone ID (`PULocationID`) and pick up date (`pickup_datetime`).
5. Joined the MTA train stations data with the TLC zone lookup dataset via a spatial join.
6. The result of Step 5 was then joined to the preprocessed and aggregated yellow taxi trip data via taxi zone ID to eventually form the column `num_stations`.
7. Residential data was also obtained, the feature `residential_density` was created, and then a spatial join with the result of Step 6 was performed to create a new feature called `residential_density`.
8. Extra features like `month` and `Borough` were added for additional information.
9. Conducted exploratory data analysis, observing the distribution of the target variable by borough, over time, etc.
10. Our two chosen models were trained and evaluated with diagnostic plots. They were also interpreted in the context of seeing where daily taxi revenue was highest.

4 Preprocessing

4.1 TLC Trip Records Data (Yellow Taxi and FHVHV)

Records had to be removed from both datasets when column values did not comply with what was deemed reasonable for our features of interest.

1. Outliers in the `total_amount` column included negative values like $-\$2265$ and $\$863380.37$ on the other extreme. These values were deemed impossible in reality and so records that had `total_amount` as or more extreme than these values were removed. In total, 5164885 records ($\approx 4\%$ of the dataset prior to this step) were removed.
2. Payment type `null` or 6 (voided trip [7]). These records were removed as we were interested in only trips that we were certain had occurred. No records were removed from this step.
3. `PULocationID` corresponding to a zone that was unknown, outside NYC or one that motor vehicles cannot access. We were only interested in taxi drivers picking up from zones within NYC or that they could actually access. A total of 581380 records ($\approx 0.5\%$ of the dataset prior to this step) were removed.

4. Filtering out records that had `tpep_pickup_datetime` in the year 2022 despite care taken to choose data from 2023 onwards. A total of 129 records ($\approx 0.0001\%$ of the dataset prior to this step) were removed.

For the NYC TLC’s HVFHV dataset, we performed only performed step 3, leading to **37581 rows removed** ($\approx 0.005\%$ of the dataset).

4.2 MTA Train Stations Data

We removed all columns in this dataset except `GTFS Latitude` and `GTFS Longitude` since we only needed the **locations** of the stations for spatial joining. Stations were expected to serve as a source of competition for yellow taxis, where daily commuters could choose to use NYC’s vast subway network or a yellow taxi to get to their desired destinations. Therefore, the expectation was that zones with higher number of subway stations would be associated with lower yellow taxi earnings. We created a data frame which would show the number of subway stations present per taxi zone by spatial joining between the `geometry` column of the TLC taxi zone geospatial data and coordinate data of each train station.

4.3 MapPLUTO: Residential and Geospatial Data

There was a large number of columns present in this dataset to highlight the various types of tax lot areas (pieces of land allocated to an owner). From this dataset, we created a feature called `residential_density`, defined as the number of residential units per building. The formula in terms of the features of the MapPLUTO dataset was: $\text{residential_density} = \frac{\text{UnitsRes}}{\text{NumBldgs}}$. We were interested in all building types except the ones that had a 0 `residential_density` (number of residential units per building), with the exception of airports. This filtering step removed 64868 records ($\approx 7.8\%$ of the dataset prior to this step). We also retained the `geometry` column for each tax lot area. This was to prepare the data for spatial joining with the TLC taxi zone geospatial data, which would allow us to calculate `residential_density` per taxi zone. `null` values in `residential_density` still had to be removed, a total of 1993 records to be exact ($\approx 5.5\%$ of the dataset prior to this step).

We chose `residential_density` as a feature to differentiate **between zones**. While this feature would not help with capturing change over time, it helps to differentiate between the 200+ taxi zones that our models are to be trained on. Our hypothesis is that zones with higher residential densities will be associated with higher daily revenue due to the higher demand potential.

5 Preliminary Analysis

5.1 Geospatial distribution of yellow taxis vs HVFHVs

Figure 1 shows the alarming difference in trip volume between HVFHVs and yellow taxis over the three year period of January 2023 to December 2025. Yellow taxis are outnumbered across practically the entire city. This stark difference is most likely attributed to the fact that taxi vendors have a yellow taxi fleet of a limited size while HVFHVs vendors such as Uber benefit from everyday citizens working for them using their own cars. It is therefore imperative for yellow taxi drivers to choose wisely where they work as they are directly competing with an ever growing number of HVFHV for passengers.

Yellow taxis are most concentrated in the airports and the borough of Manhattan. A point of concern is that HVFHVs heavily outnumber yellow taxis at the airports. We expect our models to predict

high earnings in the airports but taxi drivers should make additional considerations with regards to the traffic and competition they would experience for passengers there.

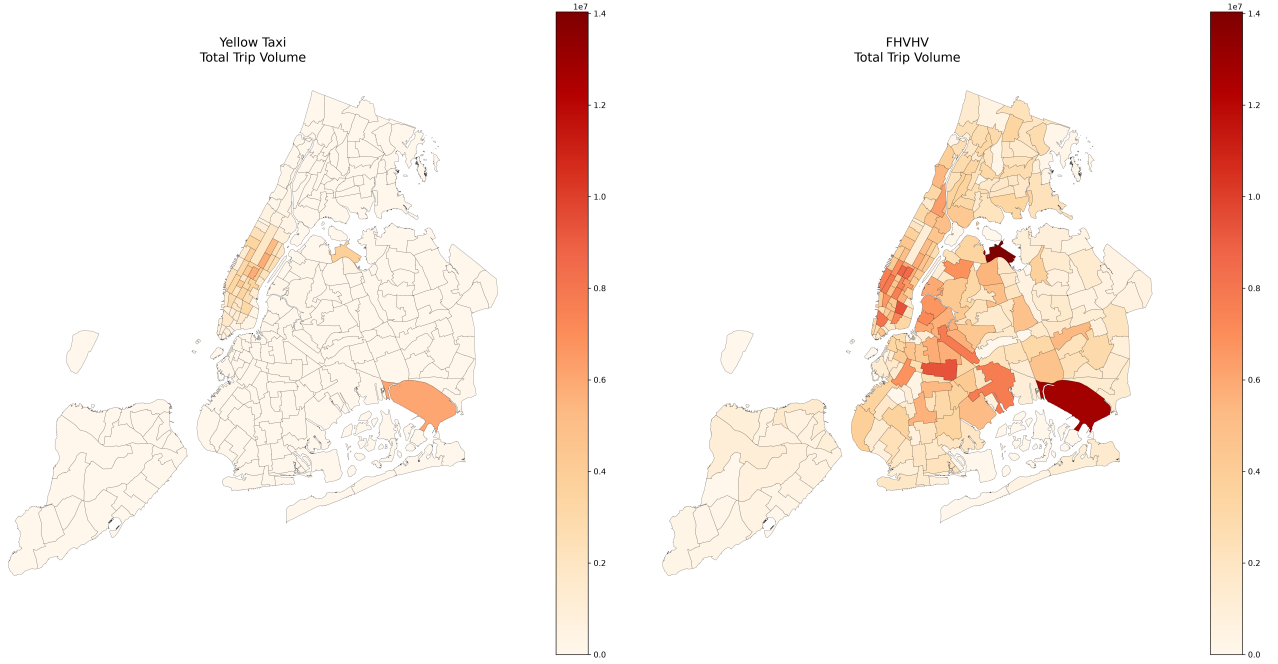


Figure 1: Spatial distribution of trips (2023-2025) between yellow taxis (left) and HVFHVs (right).

5.2 Cyclical patterns of taxi earnings

While we may have certain external factors affecting taxi earnings, our models still needed a concept of time and "progression" of taxi earnings throughout the year, to account for seasonal fluctuations due to things such as holidays, the effects of weekends vs weekdays (which affect demand for rides) among other things. Figure 2 shows a very clear seasonal pattern to yellow taxi earnings and so our model should be given this information *in addition* to external data related to subway stations and residential density. Figure 3 shows a zoom-up of a part of the plot from Figure 2, where the weekly seasonality is more clear. We expect to make use of this finding in our models where we will pass in features like `month` and `day_of_week` to help them capture these periodic patterns.

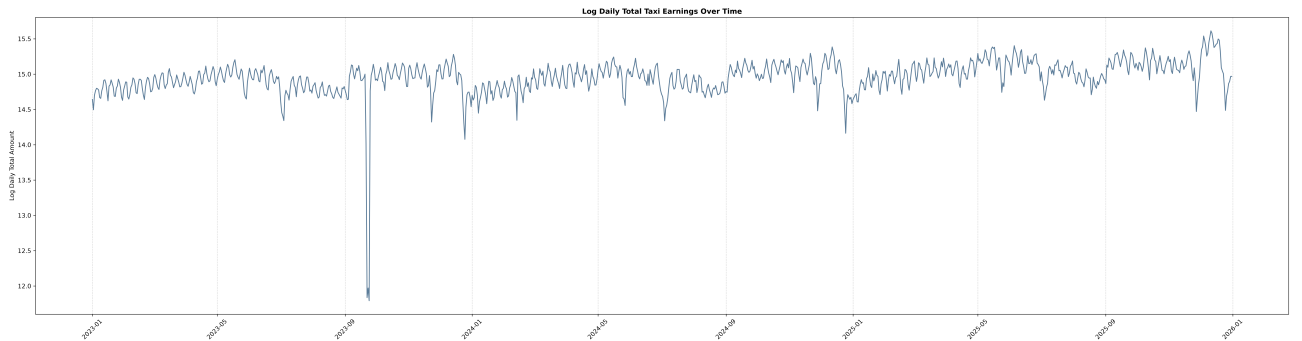


Figure 2: Total taxi earnings over 2023-2025 (logarithmic scale)

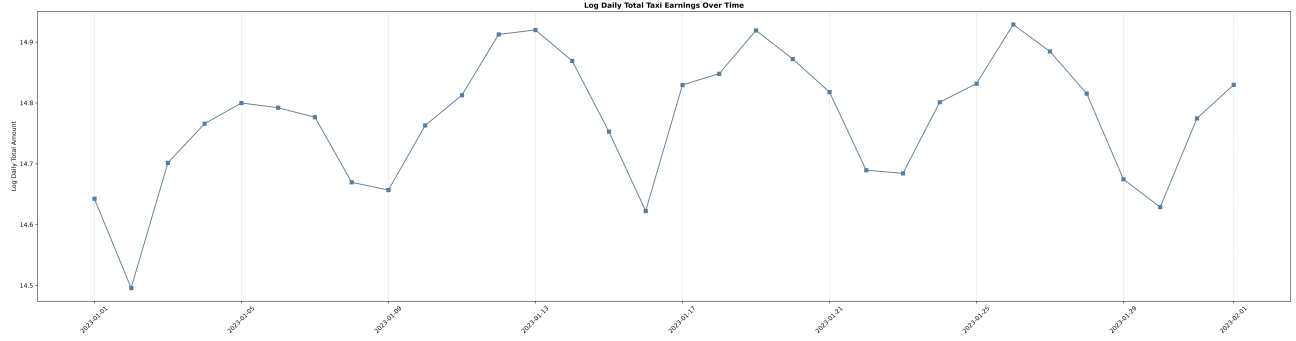


Figure 3: Zoomed view of daily total taxi revenue over a month (logarithmic scale)

5.3 Borough-specific trends

Figure 4 shows some notable differences in daily revenue between the various boroughs of NYC which we could use to our advantage in modelling. Namely, we see that taxi earnings in Manhattan over the 2023-2025 period far exceed that of the other boroughs generally speaking, with even the lower quartile of the distribution exceeding the upper fence of the earnings from the other boroughs. Our models would likely pick up this pattern and preferentially predict higher earnings in zones within Manhattan.

We also see a very compressed distribution for Newark International Airport (EWR). This is likely because EWR had much fewer records in the training data, which itself might be a reflection of the low demand for yellow taxis in that area.

When we look to Figure 5, where total number of taxi trips by borough is shown on a **logarithmic scale**, we see extremely stark differences in where yellow taxis tend to go to for passengers. Considering the vertical axis is logarithmic, Manhattan has about **10 times more** taxi trips where passengers are picked up from there compared to Queens, the second most common borough. Overall, we expect borough to be a relatively significant predictor of daily taxi revenue for a particular zone, especially if the zone is located in Manhattan.

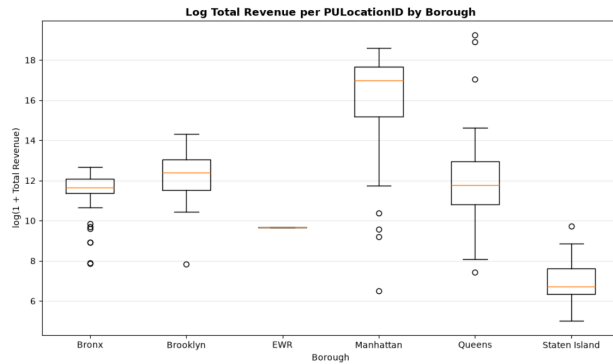


Figure 4: Log total revenue per Borough (2023-2025)

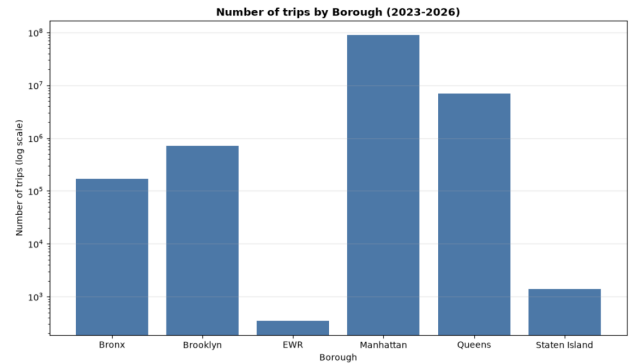


Figure 5: Trip distribution per Borough (2023-2025)

6 Modelling

We trained and analysed the performance of two models using the **mean squared error** metric. These models were chosen on the basis that our data is likely to have **temporal dependence**, or autocorrelation, where residuals are not truly independent of each other because revenue earned in a zone in the past affects today's.

HV FHV competition was simulated using a particular feature called **trip_volume** which was the number of total pickups for HV FHVs on a given day. While this information would not be available at prediction time (since the TLC takes time to upload new data for a given month), we assume that in deployment, our models would be using accurately forecasted data on the number of expected pickups on a given day by HV FHVs. To maintain focus on our research question, we avoided creating our own trip volume forecasting model but expect taxi vendors or third party companies to provide that software.

Regarding features, we used the **log transform** of **total_amount** and **trip_volume** since these features were found to be heavily right skewed. Additionally, **no subsampling was performed** on the dataset.

6.1 Generalized least squares regression (GLS)

We chose GLS as our first model because it is a linear model that could handle **autocorrelation in the residuals** [9][1]. This was important because our data had a temporal component to it, meaning the earnings of yesterday or last week could influence the earnings today. Therefore, ordinary least squares (OLS) regression would not be suitable as this data violated the assumption of independence in the residuals. This also means that if we still find the residuals of our GLS model to be autocorrelated, we should take caution with interpreting confidence intervals and standard errors.

6.2 Random Forest Regression (RFR)

RFR is a tree-based ensemble model that builds multiple random trees on samples of the training data via a technique called **bagging** [2]. This technique lends to the model having a lower variance than a single typical decision tree. Since we wanted non-linear relationships between our predictors and the total revenue to be captured where possible, RFR was expected to be a natural improvement over our GLS regression model. Furthermore, RFR does not require any assumption of independence between observations. This is critical for the temporal data we have been analysing where the revenue earned on one day is likely to be influenced by what was earned yesterday.

To minimise overfitting, we performed **time-aware cross validation** (which respects the chronological order of the data) using **scikit-learn's TimeSeriesSplit** [8] on the RFR model, yielding an optimal parameter set of **n_estimators = 100** trees and **max_depth = 20**. We did not optimise any other parameters due to processing time constraints.

7 Results and Discussion

7.1 Metrics

Table 2 shows a lower MSE for the RFR (1.1687 vs 1.6097), suggesting that non linear relationships do exist which the GLS regression model failed to capture. The results also show considerable overfitting for both models, suggesting generalisation needs improvement.

Model	Val MSE	Test MSE
GLS Regression	1.1167	1.6097
Random Forest Regressor	0.5032	1.1687

Table 2: Test performance of the regression models.

7.2 GLS Regression Model Findings

Our model summary showed a coefficient of -0.1491 for the predictor `has_station`. The negative sign is most notable because it supported our initial hypothesis prior to our analysis that the presence of a train station in a taxi zone would act as a source of competition, therefore threatening yellow taxis' earnings. However, the effect is still modest and should be taken with skepticism. We also got a coefficient of 3.8101 for `isAirport` which was consistent with our findings in the data exploration phase where we found airports to be a hot spot for yellow taxis to be concentrated. Specifically, taxis who pick up passengers from airports earn $e^{3.8101} \approx 45.2$ times more revenue daily than taxis who pick up passengers from elsewhere. Additionally, the borough of a zone was found to be about as significant as whether a zone is an airport. This is likely because airports have a naturally high demand for riders (travellers coming to or from the airport). The comparatively higher earnings in Manhattan also explain the negative coefficients of the four one hot variables corresponding to the other four boroughs. For example, taxis picking up passengers from the Bronx earn $e^{-3.4572} \approx 3.2\%$ the daily revenue that a taxi would earn from picking up passengers from Manhattan for the day.

Residential density, day of week and month were found to be the **least** important factors in deciding taxi earnings. This insinuates a more complex seasonal pattern and the fact that population-dense boroughs like Queens and Brooklyn have had historically low pickups by yellow taxis relative to Manhattan or the airports.

Further interpretation of this model, particularly using quantities that involve standard errors **will not be accurate**. Figure 7 shows that the residuals in the data do not follow a normal distribution, which means hypothesis tests and confidence intervals will be unreliable. On the other hand, Figure 6 shows the residual distribution centred at 0, insinuating that the model has not missed any systematic pattern, though this should be taken with caution due to the model being slightly mis-specified.

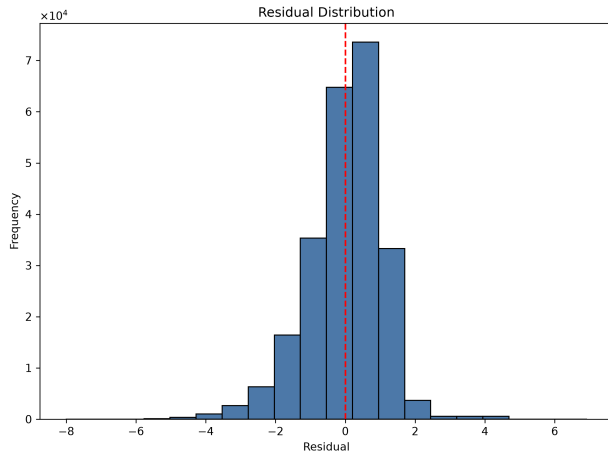


Figure 6: Distribution of residuals from GLS

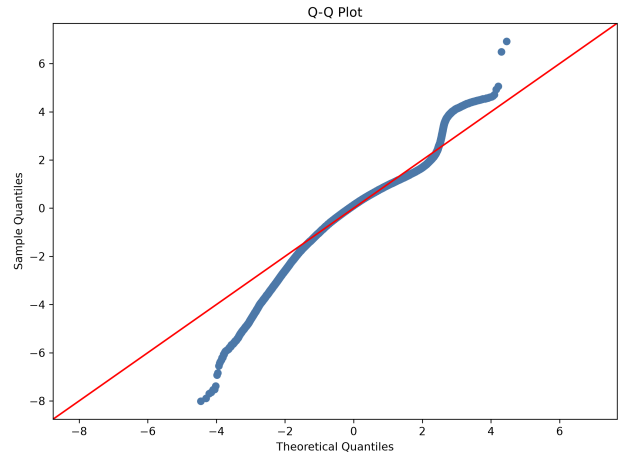


Figure 7: Q-Q plot of GLS residuals

7.3 Random Forest Regressor Findings

Figure 8 shows an almost polar opposite contrast to the features that the RFR model regarded as most informative compared to the GLS regressor. However, we see that the model regarded it as most important whether a taxi zone was within Manhattan and to a much lesser extent, also recognised the impact of taxis picking passengers from airports. Figure 9 additionally shows that the RFR model has missed a systematic trend in the 2026 data since the distribution is not centred at 0. This indicates clear room for improvement for the RFR model, which taxi vendors could look into improving with their own specific datasets.

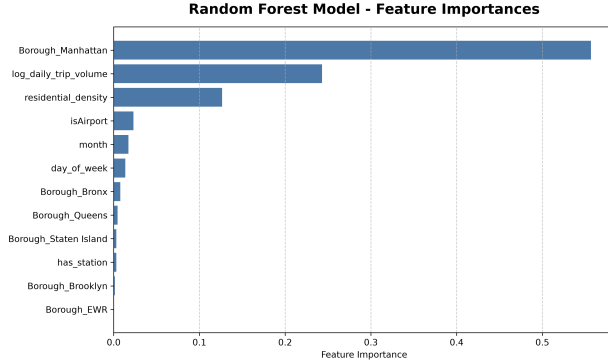


Figure 8: Feature importance from RFR

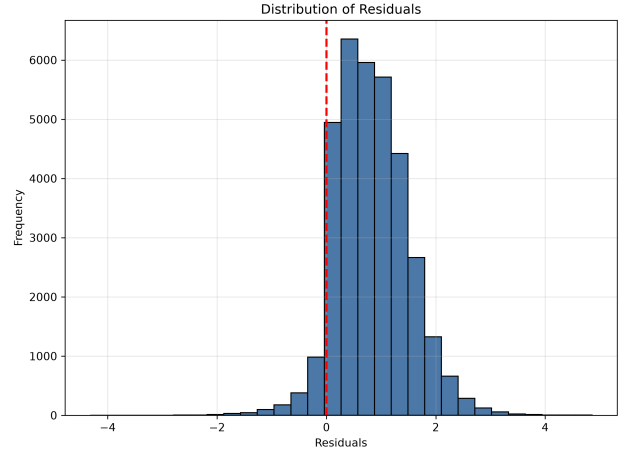


Figure 9: Distribution of residuals from RFR

8 Conclusion and Recommendations

Overall, from our analysis, we make a few conclusions and recommendations:

1. Yellow taxi numbers are very limited compared to those of HVFHVs. This makes it particularly important for yellow taxi drivers to focus their efforts on getting passengers from zones with existing high demand like Manhattan.
2. With regards to airports, our analysis shows the opportunity for high earnings there. However, drivers should be wary of the high numbers of HVFHVs in that area (as evident by Figure 1) which may cause long queues and make it less efficient to earn money there compared to less crowded areas such as the borough of Queens.
3. If there is saturation in taxis and transport vehicles in Manhattan, taxi drivers can redirect to Queens, which our analysis shows is promising in its earnings and demand, being second only to Manhattan among the boroughs.
4. Another critical recommendation to make is for taxi vendors to seek ways to improve the forecasting accuracy of our existing models. We believe RFR to be the most promising model to focus on.
5. Lastly, taxi vendors should work towards developing or acquiring a model for accurately forecasting trip volume. Our findings show that HVFHV trip volume follows a fairly predictable weekly seasonal pattern, and so vendors could look into using models such as ARIMA which are specifically designed for autocorrelated and seasonal data.

References

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